

Why Machine Learning in Physics?

Lecture 02 — PHYG004, 2026 Spring

Prof. Young Woo Choi

Department of Physics, Sogang University

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The AI Moment We're In

The Hype

- "AGI is near" headlines
- AI investment bubble (\$300B+ in 2024–25)
- Generated content flooding the internet
- Exaggerated benchmark competitions

The Real Progress

- Code generation that actually works (Copilot, Claude Code)
- Agentic systems: autonomous experiment design
- Scientific breakthroughs (next slide)

We need to separate **signal from noise** — especially as physicists.

AI for Science: Breakthroughs That Changed the Game



AlphaFold (2020, 2024)

Protein structure prediction
— solved a 50-year grand
challenge

**200M+ structures
predicted**

Structural Biology



GNoME (2023)

Graph NN trained on 89M
DFT calculations

2.2M new stable crystals

Materials Science



FermiNet / Psiformer (2020–24)

NN wavefunctions
surpassing gold-standard
quantum chemistry

Beyond CCSD(T) accuracy

Quantum Chemistry

These are not incremental improvements — they are **paradigm shifts**.

Three Paradigms of AI for Science



1. Pattern Recognition

Data → Pattern →
Classification



2. Surrogate Model

Expensive calc → Fast NN
proxy

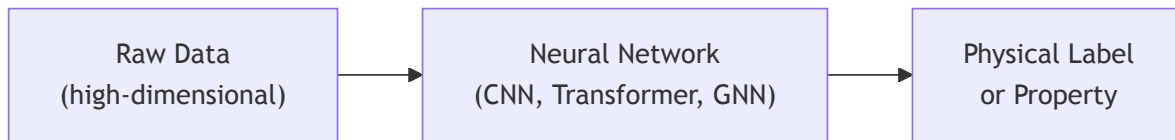


3. Agent

Observe → Decide → Act

This course focuses on **1 and 2**. Paradigm 3 is emerging rapidly.

Paradigm 1: Pattern Recognition



Input

Spin configurations

→

Detector signals

→

Spectra

→

Output

Phase: ordered / disordered

Event: signal / background

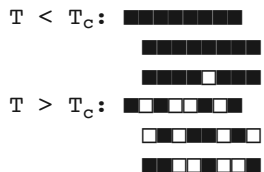
Material class

Key idea: The NN extracts patterns that may be invisible to human analysis or lack a known order parameter.

Pattern Recognition in Action

Example 1: Phase Transition Classification

- Carrasquilla & Melko, *Nature Physics* (2017)
- Input: 2D Ising MC spin configurations (32×32)
- Model: Simple CNN
- Output: Ordered ($T < T_c$) vs Disordered ($T > T_c$)



NN finds T_c accurately — even for topological phases!

Example 2: Gravitational Wave Detection

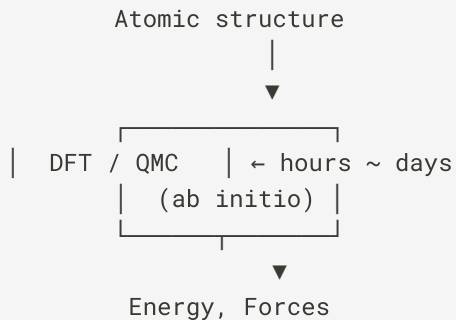
- George & Huerta, *Phys. Lett. B* (2018)
- Input: LIGO time-series strain data
- Model: Deep CNN
- Output: Signal present / absent

Matched filtering sensitivity, but **real-time** processing

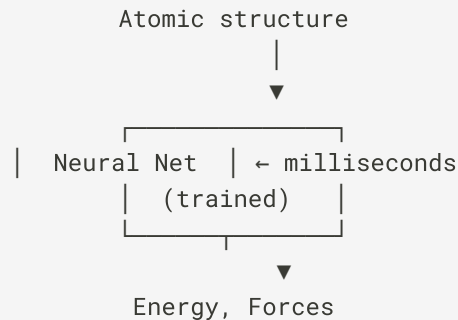
Both examples use CNN — we build the first one in **L06**.

Paradigm 2: Surrogate Model

Traditional



ML Surrogate



Speed: 10^3 – 10^6 × faster | **Accuracy:** DFT-level or beyond | **Scale:** millions of atoms

Surrogate Models in Action

1. Machine Learning Interatomic Potentials (MLIP)

L13

- NequIP, MACE: E(3)-equivariant GNN maps atomic structure $\rightarrow$ energy/forces
- DFT accuracy + MD speed $\rightarrow$ million-atom simulations

2. Neural Network Wavefunctions

L03, L08

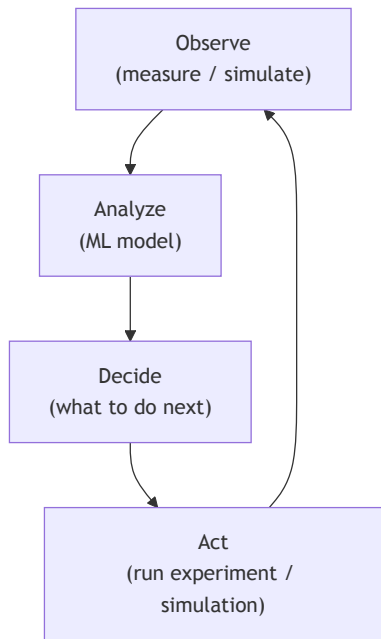
- FermiNet, Psiformer: NN represents $\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N)$
- Variational Monte Carlo training $\rightarrow$ beyond CCSD(T)

3. Learning the Hamiltonian Directly

L13

- DeepH: atomic structure $\rightarrow$ DFT Hamiltonian matrix
- Electronic structure of 10,000+ atoms without SCF iteration

Paradigm 3: Agent



- ML decides **what experiment/calculation to run next**
- Active learning: explore regions of highest uncertainty
- Bayesian optimization: minimize objective with fewest evaluations

Agent Paradigm: Examples

1. Active Learning for Training Data

- Problem: MLIP training needs DFT data, but DFT is expensive
- Solution: auto-select structures where NN uncertainty is highest → run DFT → add to training set
- Result: **10× fewer DFT calculations** needed

2. Self-Driving Laboratories

- Acceleration Consortium (U of T), A-Lab (LBNL)
- ML proposes synthesis conditions → robot synthesizes → characterize → ML updates
- 24/7 autonomous operation for accelerated materials discovery

In this course, we focus on Paradigms 1 & 2. But many term projects can incorporate active learning ideas.

Course Roadmap

Lecture	Topic	Block	Paradigm Connection
L01	JAX, Autodiff, Physics-vs-ML	Foundations	Foundation
L02	Why ML for Physics (this lecture)	Foundations	Survey
L03	NN Foundations: Backprop, Activations	Foundations	Foundation
L04	Classification & Information Theory	Foundations	Pattern Recognition
L05	MLP Deep Dive: Optimization, Overfitting	Foundations	Foundation
L06	CNNs: LeNet to Phase Classification	Foundations	Pattern Recognition
L07	CNN Physics Apps + Transfer Learning	Foundations	Pattern Recognition
L08	Autodiff & PINNs (Full Session)	Foundations	Surrogate
L09	Differentiable Physics: ODE & Inverse Problems	Foundations	Surrogate

Part 2

Physics Modeling vs ML Modeling

How Physicists Build Models

Fundamental principles (Newton, Schrödinger, Maxwell)

↓ symmetry → conservation laws

Mathematical structure (Hamiltonian, Lagrangian)

↓ approximation → mean-field, perturbation

Tractable equations (Hartree-Fock, Landau theory)

↓ dimensional reduction → order parameter

Predictions (phase diagram, spectrum)

Symmetry

Approximation

Dim. Reduction

The Physicist's Toolkit

Symmetry → Reduce the space

- Rotational symmetry → angular momentum quantum numbers ℓ, m → block-diagonalize Hilbert space
- Translation symmetry → Bloch's theorem, reduction to k -space

Approximation → Make it solvable

- Mean-field: many-body → effective single-body (e.g., Hartree-Fock)
- Perturbation theory: expand in small parameter (e.g., $e^2/\hbar c \approx 1/137$)
- Variational: trial wavefunction + energy minimization (e.g., VMC)

Dimensional Reduction → Capture the essence

- Order parameter: 10^{23} spins → magnetization m
- Effective theory: integrate out high-energy DOF → low-energy EFT
- Renormalization group: remove irrelevant DOF scale by scale

Where Physics Modeling Breaks Down

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) \in \mathbb{C}^{3N\text{-dimensional space}}$$

N (electrons)	Dimension ($3N$)	Grid points (10/axis)	Feasibility
2	6	10^6	✓ tractable
10	30	10^{30}	✗
50	150	10^{150}	✗ universe has $\sim 10^{80}$ atoms
100	300	10^{300}	✗✗✗

This is why we need ML. Traditional basis expansions fail exponentially. Neural networks offer a way out.

How ML Builds Models: Function Approximation

Given data $\{(x_i, y_i)\}_{i=1}^N$, find f_θ such that $f_\theta(x_i) \approx y_i$

- f_θ : function parameterized by θ (neural network)
- Objective: minimize loss $\mathcal{L}(\theta) = \frac{1}{N} \sum_i |f_\theta(x_i) - y_i|^2$
- Update rule: $\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}$

Universal Approximation Theorem (Cybenko 1989, Hornik 1991): A single hidden layer NN with sufficient width can approximate any continuous function on a compact domain to arbitrary accuracy.

But this theorem says nothing about **how many parameters** or **how much data** you need.

The Curse of Dimensionality

d -dimensional unit cube $[0, 1]^d$, grid spacing ϵ :

$$N_{\text{grid}} = \left(\frac{1}{\epsilon}\right)^d$$

With $\epsilon = 0.1$ (10 points per axis):

d	N_{grid}	
3	10^3	✓ Easy
10	10^{10}	⚠ Expensive
30	10^{30}	✗ Impossible
100	10^{100}	✗✗ Absurd

1D: • • • • • (10)

2D: • • •

• • •

• • • (100)

3D: [cube of 1000 points]



continues **exponentially...**

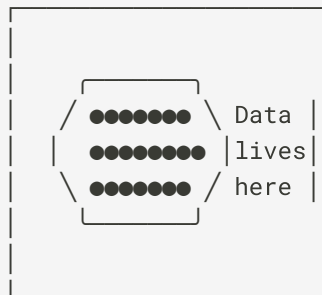
Why ML Works: Data Lives on Low-Dimensional Manifolds

Physical data occupies a **tiny, structured subset** of the full high-dimensional space.

Ising Model Example:

- Configuration space: $\{-1, +1\}^{L \times L} \rightarrow$ for $L = 32$: 2^{1024} possible states
- Thermodynamically relevant: only those weighted by $p(\sigma) \propto e^{-\beta H(\sigma)}$
- Effective dimension: (T, m) — a **2-dimensional manifold**

Full space (2^{1024} states)



Neural networks are **implicitly learning this manifold structure**.

ML Training $\equiv$ Physical Optimization

Physics (Variational Principle)

$$E[\Psi_\theta] = \frac{\langle \Psi_\theta | \hat{H} | \Psi_\theta \rangle}{\langle \Psi_\theta | \Psi_\theta \rangle} \geq E_0$$

- Trial wavefunction Ψ_θ
- Minimize energy expectation
- $\theta \leftarrow \theta - \eta \nabla_\theta E[\Psi_\theta]$

ML (Loss Minimization)

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(f_\theta(x_i), y_i)$$

- Neural network f_θ
- Minimize loss function
- $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}(\theta)$

Identical mathematical structure: parameter space + objective function + gradient descent.

Inductive Bias: The Key to Good ML

Inductive bias: The set of assumptions a learning algorithm uses to predict outputs for unseen inputs. *"What shape of function do you expect before seeing any data?"*



Linear

"straight line"



Polynomial

"smooth curve"



Periodic

"repeating pattern"

- Same data, different assumptions → completely different predictions
- No Free Lunch Theorem: **no learning without assumptions**
- **The better the physics, the stronger the inductive bias**

For physicists: inductive bias = **symmetry, conservation laws, locality**

Physics Priors → Neural Network Architectures

Physical Prior	Math	NN Architecture	Lecture
Translation symmetry	$f(x + a) = f(x)$	CNN (shared weights)	L06
Permutation symmetry	$f(P\mathbf{x}) = f(\mathbf{x})$	GNN (message passing)	L11
Rotation/reflection	$f(R\mathbf{x}) = D(R)f(\mathbf{x})$	E(3)-equivariant NN	L12
Energy conservation	$dE/dt = 0$	Hamiltonian NN	—
Fermion antisymmetry	$\Psi(\dots i \dots j \dots) = -\Psi(\dots j \dots i \dots)$	Determinant layer	L03
Probability normalization	$\int p(x)dx = 1$	Normalizing flow	L15

Hamiltonian NN is a natural extension of autodiff (L08/L09); not covered as a standalone lecture in this curriculum — see L08 PINN and L09 differentiable physics as entry points.

Example: CNN Encodes Translation Symmetry

MLP (no symmetry)

- Input: flatten 32×32 spin lattice to 1D vector
- Same domain pattern at different positions $\rightarrow$ different input $\rightarrow$ different output

[■□□□□...] $\rightarrow$ MLP $\rightarrow$ "ordered"

[□■□□□...] $\rightarrow$ MLP $\rightarrow$ "???" $\leftarrow$ different input!

"MLP treats domains at (3,5) and (20,15) as completely different patterns"

CNN (translation equivariant)

- Same filter scans all positions $\rightarrow$ position-independent features

[■□□□□...] $\rightarrow$ CNN $\rightarrow$ "ordered"

[□■□□□...] $\rightarrow$ CNN $\rightarrow$ "ordered" ✓

"CNN detects the domain regardless of where it appears"

CNN's weight sharing = translation invariance **built into** the architecture.

The Power of Encoding Symmetry: 1000× Less Data

Molecular/crystal energy is invariant under rotation, reflection, translation:

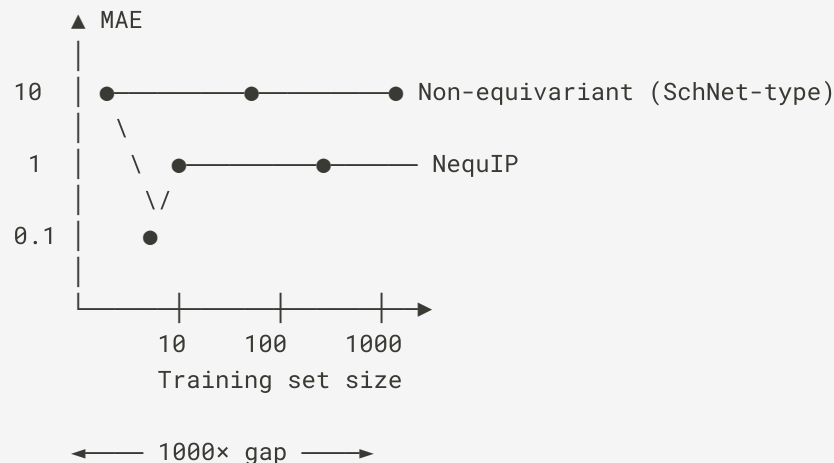
$$E(R\mathbf{x} + \mathbf{t}) = E(\mathbf{x})$$

Forces are equivariant under rotation:

$$\mathbf{F}(R\mathbf{x}) = R \cdot \mathbf{F}(\mathbf{x})$$

NequIP: encodes this symmetry exactly in GNN architecture

Energy MAE (meV/atom) vs Training data



Both models output invariant scalar energy; the gap reflects equivariant vs invariant intermediate features — NequIP uses equivariant intermediate representations, SchNet-type uses invariant intermediates.

Two Ways to Handle Symmetry

Data Augmentation (brute force)

- Method: duplicate training data with rotations/reflections
-  Simple to implement, works with any model
-  Training data $\times |G|$, training time $\times |G|$
-  Symmetry learned only **approximately**
-  Continuous symmetries ($SO(3)$) impossible to cover exactly

Equivariant Architecture (by design)

- Method: each NN layer commutes with symmetry transforms
-  No additional data needed
-  Symmetry guaranteed **exactly**
-  Generalization is automatic
-  Architecture design is more complex

Lesson: When you know the symmetry, encode it in the architecture. Data augmentation is a fallback, not a solution.

Translating Physics Problems into ML Tasks

Given a physics problem, **what kind of ML task is it?**

- 1. Approximation** → "Represent a complex function"
- 2. Inference** → "Infer hidden variables from observations"
- 3. Sampling** → "Generate samples from a distribution"

Every physics-ML project starts by identifying which task type applies.

Task Type 1: Approximation — Represent Complex Functions

Learn a parametric function $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ from data or by optimization.

In physics: represent complex functions where traditional basis expansions fail

Physics Problem	Input $\mathcal{X}$	Output $\mathcal{Y}$	Model
Interatomic potential	Atomic positions $\{\mathbf{r}_i\}$	Energy E , Forces $\mathbf{F}$	NequIP, MACE
Quantum wavefunction	Electron coords $\mathbf{r}_1, \dots, \mathbf{r}_N$	$\Psi(\mathbf{r})$	FermiNet
DFT Hamiltonian	Crystal structure	H_{ij} matrix elements	DeepH

Bottleneck solved: Basis expansion fails exponentially; NNs provide compact, flexible representations.

Task Type 2: Inference — From Observations to Hidden Variables

Given observations $\mathbf{d}$, estimate hidden parameters ϕ :

$$P(\phi | \mathbf{d}) \propto P(\mathbf{d} | \phi) P(\phi)$$

Physics Problem	Observations $\mathbf{d}$	Hidden ϕ	ML Method
Quantum state tomography	Measurement outcomes	Density matrix ρ	Generative NN
Phase classification	Spin configurations	Phase label, T_c	CNN classifier
Simulation-based inference	Detector data	Theory parameters	Neural posterior (L19)

Task Type 3: Sampling — Generate Configurations from Distributions

Generate independent samples $\mathbf{x} \sim p(\mathbf{x})$ where $p(\mathbf{x}) = \frac{1}{Z} e^{-\beta E(\mathbf{x})}$

Traditional: MCMC (Metropolis, HMC) → **autocorrelation, critical slowing down** near T_c

ML solution:

- **Normalizing flow** (L15): simple distribution → complex distribution, **one-shot** independent samples
- **Flow matching** (L16): learn continuous velocity field, simulation-free training
- **Diffusion model** (L17): start from noise → gradually generate structure
- **Energy-based model** (L18): learn $\nabla \log p$ (score) → Langevin sampling

Examples:

Three ML Task Types: Summary

ML Task	Physics Question	What NN Does	Key Example	Lectures
Approximation	Represent complex function	$f_{\theta}(\mathbf{x}) \approx y$	MLIP, NQS	L03–L09, L11–L13
Inference	Infer hidden variables	$P(\phi \mid \mathbf{d})$	QST, SBI	L04, L10, L19
Sampling	Generate from distribution	$\mathbf{x} \sim p(\mathbf{x})$	Flow, Diffusion	L14–L18

Many real problems **combine** these tasks. E.g., Neural network quantum states = Approximation + Sampling (VMC).

Summary & What's Next

Today's Key Messages:

1. **AI for Science** works through three paradigms: pattern recognition, surrogate modeling, agents
2. **Physics modeling vs ML**: both solve optimization in parameter space — identical mathematical structure
3. **Inductive bias** is the bridge: encoding symmetry into NN architectures yields dramatic improvements
4. **Three ML task types** — approximation, inference, sampling — framework for any physics → ML translation

Next:

- **Now**: Hands-on — Gradient descent on potential energy surfaces (JAX notebook)
- **L03** (next week): Neural Networks Basics — linear models → MLP → backpropagation

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